# Solar PV Installation Roadmap

Step-by-Step Execution Protocol & SANS 10142-1 Compliance Workflow

Standard Scope **3-Bed Residence (RSA)**
**STAGE 1** Pre-Install & SSEG
**STAGE 2** Mechanical Rail & Array
**STAGE 3** Electrical & Inverter
**STAGE 4** Commissioning & CoC

# Stage 1: Pre-Installation Audit & Municipal Permitting

01

### Site Assessment & Structural Roof Audit

Day -7 to -3

Inspect physical roof integrity, shading vectors, and local Sub-Distribution (DB) board layout to verify mechanical and electrical compatibility.

- Determine roof tile/sheet type (Concrete tile, IBR, or Corrugated) and rafter spacing.
- Perform azimuth and tilt measurement (optimal SA azimuth: True North ± 15°, 25°–30° tilt).
- Identify main DB supply (Single-phase 230V / Three-phase 400V) and evaluate space for extra breakers.

02

### Municipal SSEG Application & Inverter Check

Day -5 to -1

Submit Small-Scale Embedded Generation (SSEG) preliminary registration to the local municipality (e.g., City Power, Tshwane, Cape Town, or Eskom).

- Confirm chosen hybrid inverter is NRS 097-2-1 type-approved for grid connection.
- Draft preliminary Single Line Diagram (SLD) indicating main isolation and reverse power blocking.
- Obtain Permission-to-Install (PTI) letter prior to mounting panels or making grid connections.

**Legal Compliance Notice:** Operating an unregistered solar PV system or operating without a valid Electrical CoC voids residential home insurance policies and violates municipal electricity supply by-laws.


# Stage 2: Mechanical Roof Mount & Solar Array Assembly

03

### Roof Anchor Bracket & Rail System Mounting

Day 1 (Morning)

Secure roof hooks directly into wooden roof trusses to form a weather-proof, wind-resistant structural foundation for the solar panels.

04

### Panel Placement, Interconnection & Array Earthing

Day 1 (Afternoon)

Mount 550W panels to rails using mid/end clamps, series-connect strings with MC4 connectors, and bond array structures to ground per SANS 10142-1.

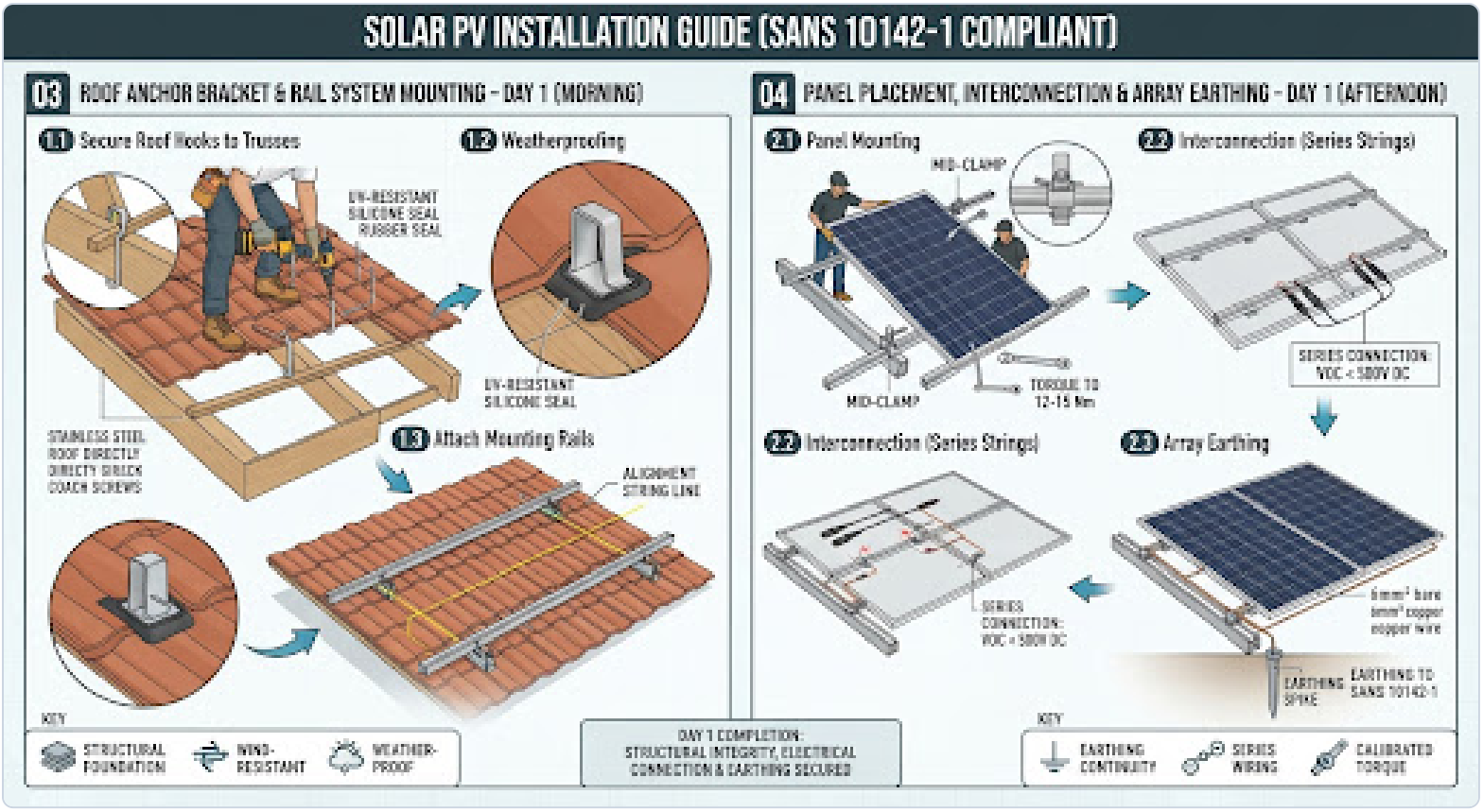



Figure 1: SANS 10142-1 compliant mounting, rail alignment, series stringing, and earthing installation protocol.

# Stage 3 & 4: Electrical Installation & Commissioning Summary

05 - 07

## Inverter, Protection Box & DB Load Splitting

Day 2

Mount 5kW hybrid inverter and 10kWh lithium battery. Wire DC switchgear (surge protection/fuses), AC changeover switch, and separate DB into essential and non-essential busbars.

08 - 09

## Testing, Electrical CoC Sign-Off & SSEG Approval

Day 3

Conduct VOC/polarity cold testing, set battery parameters, issue statutory Electrical Certificate of Compliance (CoC), and submit municipal SSEG commissioning report.